

Docblock

Provides a comprehensive docblock explaining the function's purpose, parameters, return value, potential exceptions, and assumptions about the input data type.

Error Handling:

Throws a more informative `InvalidArgumentException` if the input array is empty or contains non-numeric values, ensuring type safety.

Type Validation:

Uses `array_is_list` and `array_filter` with `is_numeric` to verify that the input array is a list of numeric values.

Meaningful Variable Names:

Employs descriptive names like `$currentValue` for improved readability.

Explanation and Enhancements:

Explains the function's logic, including the use of `array_is_list`, `array_filter`, and `array_reduce`. It also highlights the use of meaningful variable names and error handling.

Provides a detailed explanation of the function's purpose and the specific steps taken to ensure type safety and readability.

Includes a section on error handling, explaining the use of `InvalidArgumentException` to provide more context when the input is invalid.

Describes the type validation process, showing how `array_is_list` and `array_filter` are used to ensure the input is a list of numeric values.

Highlights the use of meaningful variable names, such as `$currentValue`, to make the code more self-explanatory.

Summarizes the function's overall behavior and the enhancements made to improve its clarity and robustness.

Time Complexity:

The dominant operation is the `foreach` loop iterating through the array. In the worst case, it executes n times, where n is the number of elements in the array.

Other operations (initialization, comparisons, and returning the result) are constant time ($O(1)$).

The overall time complexity of the function is $O(n)$, which is linear.

It scales proportionally with the size of the input array.

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